

Application-Driven FSR Design in Silicon Photonics

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Course: edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis

1. Introduction

1.1 Background & Motivation

Silicon photonics enables high-density Photonic Integrated Circuits (PICs) using standard CMOS manufacturing processes. Critical integrated optical elements—such as dense wavelength-division multiplexers (DWDM), bio-sensors, resonant filters, and single-mode laser, depend on precise control over phase propagation, modal dispersion, and Free Spectral Range.

1.2 Design Objectives

This project presents an extended **application-driven, geometry-modulated, and process-variability-tested methodology** for Mach-Zehnder Interferometer (MZI) circuits. The objectives are three-fold:

1. **Application-Driven FSR Unbalanced MZIs:** Design and evaluate four unbalanced MZIs tailored for system-level application domains: DWDM (0.4nm-1.6nm), sensors (2nm-10nm), Micro-Ring Resonator (5nm-20nm), and Single-Mode Laser (30nm-50nm).
2. **Manufacturing Repeatability Evaluation:** Duplicate the four application-driven MZI designs into two identical rows (Top and Bottom rows) across the chip layout to quantify fabrication tolerances and spatial repeatability.
3. **Width-Asymmetric Balanced MZI Test Suite:** Implement an equal-arm-length ($L_1=L_2$) balanced MZI with asymmetric waveguide widths ($W_1 \neq W_2$) placed on the right side of the chip to induce phase interference purely via modal dispersion (Δn_{eff}), alongside a symmetric balanced control device ($W_1=W_2$).

All structures were designed in KLayout and validated via Lumerical INTERCONNECT circuit simulations.

2. Theory & Mathematical Formulation

2.1 Waveguide Dispersion & Group Index

The optical phase shift ϕ inside a waveguide of length L is:

$$\phi = \beta L = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda) L$$

For MZI with arm lengths L_1, L_2 and effective index $n_{\text{eff}1}, n_{\text{eff}2}$, the phase difference $\Delta \phi(\lambda)$ between two paths is:

$$\Delta \phi(\lambda) = \frac{2\pi}{\lambda} [n_{\text{eff}1}(\lambda) L_1 - n_{\text{eff}2}(\lambda) L_2]$$

For FSR:

$$\Delta \phi(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda) \Delta L \implies \text{FSR} = \frac{\lambda_0^2}{n_g(\lambda_0) \cdot \Delta L}$$

3. Design Specifications & Numerical Simulation Results

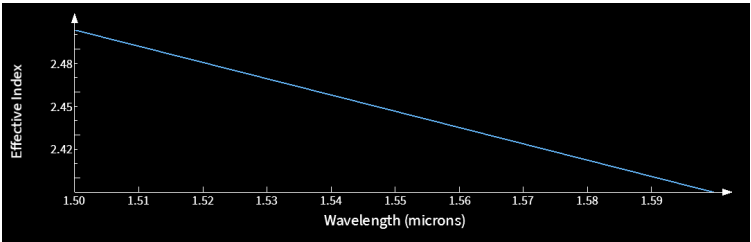
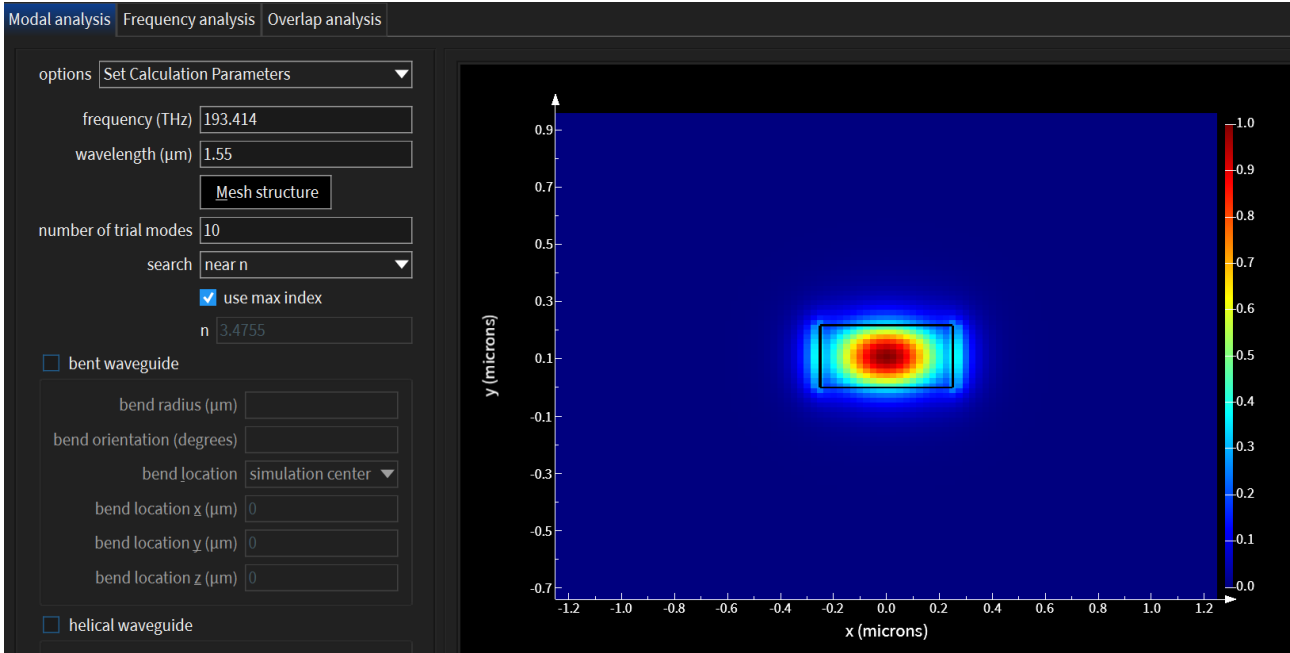
3.1 Comprehensive Parameter & Application Matrix

Waveguide dimension : width=500nm, height=220nm

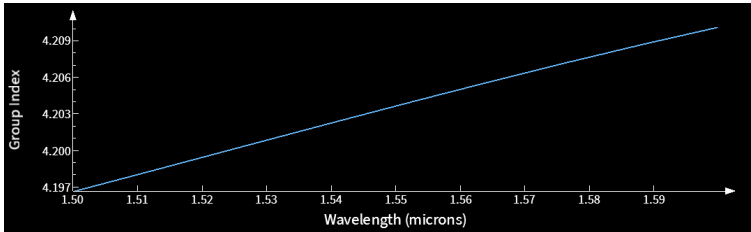
Source: quasi-TE, 1550nm

	MZI_DWDM	MZI_Sensor	MZI_MRR	MZI_Laser
ΔL	356.83um	108.4um	57.66um	16.64um
Waveguide width	500nm	500nm	500nm	500nm
FSR	1.6nm	5.24nm	24nm	30nm

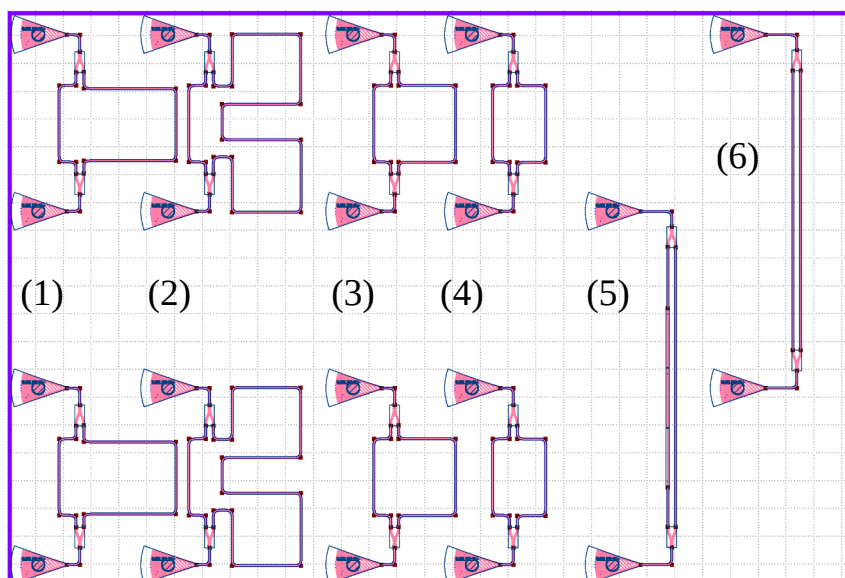
mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ex)	waveguide TE/TM fraction (%)	effective area (μm ²)
1	2.446821+1.249901e-09i	1.55	0.00044009	4.203676+2.666270e-09i	98	76.12 / 81.96	0.190575
2	1.768817+8.502993e-10i	1.55	0.00029939	3.720462+4.417996e-09i	4	68.93 / 88.89	0.368811



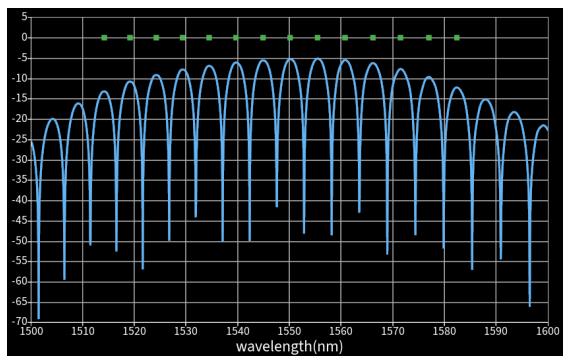
Effective index vs wavelength



Group index vs wavelength

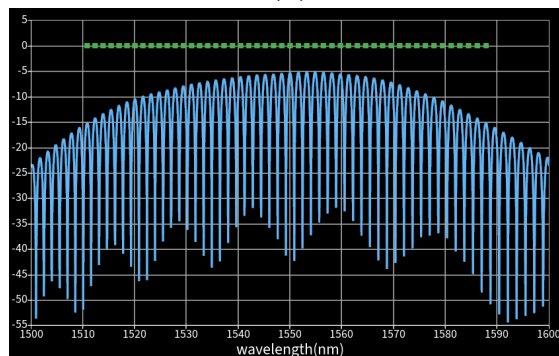


(1)



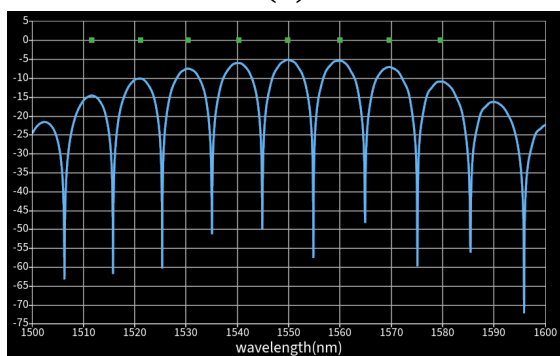
FSR ≈ 5.24 nm ($\Delta L \approx 108.4$ nm)

(2)



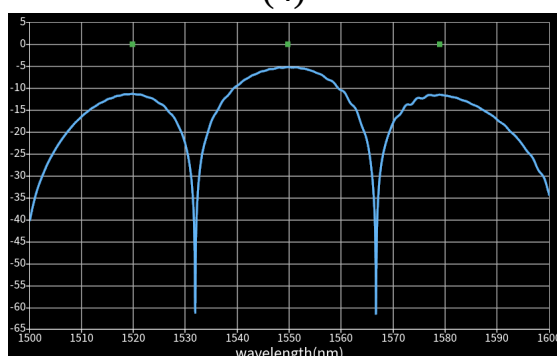
FSR ≈ 1.6 nm ($\Delta L \approx 356.83$ nm)

(3)

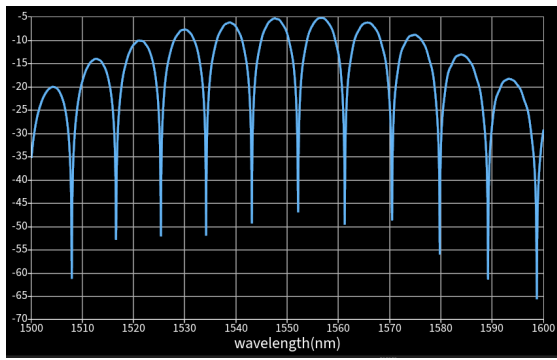


FSR ≈ 24 nm ($\Delta L \approx 57.66$ nm)

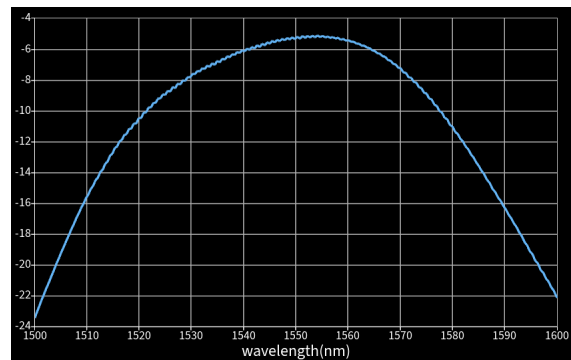
(4)



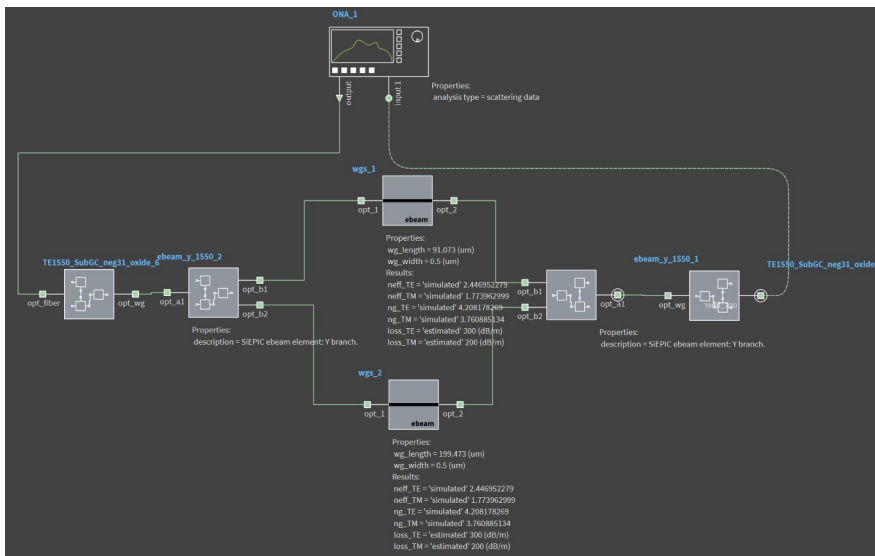
FSR ≈ 30 nm ($\Delta L \approx 16.46$ nm)



$W1 = 0.5\mu\text{m}, W2 = 0.5\mu\text{m}$



$W1 = 0.5\mu\text{m}, W2 = 0.5\mu\text{m}$



Schematic and simulation setup in Lumerical Interconnect

4. Experimental Data *(To be updated post-fabrication)*

5. Analysis *(To be updated post-fabrication)*

6. Conclusion